

Access control automation with Python

1. Project overview

In this project, I developed a Python-based solution to automate the maintenance of an access control list (ACL) containing approved IP addresses. The objective was to ensure that employees who no longer required access to restricted resources were automatically removed from the allow list. The solution compares an allow list against a remove list. Identifies unauthorized IP addresses, removes them from the approved access list, and updates the file accordingly.

2. Objectives

- read and process access control data stored in text files
- Automate the removal of unauthorized IP addresses
- Reduce the risk of human error in access management
- Demonstrate basic Python scripting skills in a cybersecurity context

3. Technologies Used

- Python
- File handling (open(), read(), write())
- Lists and string manipulation
- Conditional statements
- Iteration (for loops)

4. Process

The algorithm performs the following actions:

- A) opens and reads the allow list file
- B) Converts the file contents into a list of IP addresses
- C) Iterates through a separate remove list
- D) Checks whether each IP address exists in the allow list.
- E) Removes matching IP addresses
- F) Converts the updated list back into string format
- G) Writes the revised allow list back to the file

5. Security relevance

Access management is a fundamental cybersecurity control. Maintaining accurate access lists helps organizations enforce the principle of least privilege and reduce the risk of unauthorized access to sensitive systems and data. This project demonstrates how simple automation can improve the consistency and efficiency of security administration tasks.

6. Key skills demonstrated

- python scripting
- Access control management
- Security process automation
- Data handling and validation
- Logical problem solving

7. Outcome

The final solution successfully automated the review and update of an IP allow list by removing addresses identified as no longer authorized. This reduced manual effort and improved the accuracy of access control management.